Pranavesh Kumar Talupuri

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Research Interests

Trustworthy machine learning, Representation learning, multimodal learning, information retrieval, and differential privacy.

Education

University of Southern California, Master of Science in Computer Science — Aug 2025 – May 2027

- GPA: 3.75 (Tentative)

Vellore Institute of Technology - AP, B.Tech. in Computer Science and Engineering — Aug 2021 – May 2025
(Specialization in Artificial Intelligence and Machine Learning)

- GPA: 9.0/10

Research Experience

Graduate Research Assistant, University of Southern California — Feb 2026 – Present

- Conducting research in machine learning, differential privacy, and privacy-preserving synthetic data generation at FoRT-ML with Prof. Sai Praneeth Karimireddy, collaborating with Mohammadamin Banayeeanzade and Tony Yang.
- Investigating privacy–utility trade-offs in synthetic agentic tool-use trajectories, including structural representations, trajectory validity, and downstream learning utility.

Machine Learning Researcher (Intern), Virginia Tech Research Forum — Jan 2025 – Jun 2025

- Developed a machine learning model for accelerated discovery of high-performance piezoelectric materials.
- Built computer vision and deep learning models for an autonomous underwater vehicle, contributing to perception for robotics applications.

Undergraduate Research Assistant / Project Lead, Vellore Institute of Technology — Oct 2022 – May 2025

- Led a team of 7 student engineers to design, build, and test a functional VTOL prototype.
- Initiated computer vision applications on the VTOL platform, including surveillance and survey use cases.
- Worked across hardware prototyping and perception software for robotics systems and applied vision research.

Publications

SGRA: Empirical Support Geometry for Nested Multimodal Embeddings
Pranavesh Kumar Talupuri
Manuscript under review at WACV 2027 (Applications Track).

From Hand-Drawn Schematics to Structural Circuit Netlists: A Deterministic Pipeline with Endpoint-Graph Wire Joining and a Human-Verified Connectivity Benchmark
Bosco Chanam, Chris Dcosta, Pranavesh Kumar Talupuri, Shwetambari Chiwhane, Ashay Kumar Singh, and Arghadeep Das.
Manuscript under review at IEEE Access.

Analysis of Acute Lymphoblastic Leukemia Detection Methods Using Deep Learning
Pranavesh Kumar Talupuri, Beebi Naseeba, Nagendra Panini Challa, Abbaraju Sai Sathwik.
2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), 2023.
doi:10.1109/ICSSAS57918.2023.10331776

Smart MCQ Generation using KeyBERT and BERT Based Models
Pranavesh Kumar Talupuri, Manjunaatt Girish Kumar, Kotha Lavanya, Gudditi Chetan, Nagendra Panini Challa.
2024 3rd Edition of IEEE Delhi Section Flagship Conference (DELCON), 2024.
doi:10.1109/DELCON64804.2024.10866615

Hybrid Time Series Forecasting with ARIMA, Echo State Networks, and LSTM
Sai Eeshwar D, Pranavesh Kumar Talupuri, Aarush Shivkumar, Beebi Naseeba.
2025 International Conference on Computing Technologies & Data Communication (ICCTDC), 2025.
doi:10.1109/ICCTDC64446.2025.11158940

Comparison of Deep Learning Models for Classification of Acute Lymphoblastic Leukemia
Pranavesh Kumar Talupuri, Nitin Bhimineni, Beebi Naseeba, E. R. Aruna, Sai Vennela Tipirneni, Nagendra Panini Challa.
Pertanika Proceedings, 1(5):93–99, 2025. doi:10.47836/pp.1.5.018

Selected Research Projects

Movement Is Not Adaptation: Auditing Intent Edits in Instruction-Following Retrieval

- Developing a transition-validity evaluation framework for instruction-following retrieval, examining whether local intent edits produce beneficial ranking changes rather than arbitrary result movement.
- Investigating adaptation gain, constraint retention, paraphrase stability, ranking movement, and query-representation movement across dense retrieval models and model scales.
- Manuscript in preparation for *ACM Transactions on Information Systems (TOIS)*.

AnytimeVLE: Retrofitting Frozen Vision–Language Indexes for Risk-Controlled Adaptive Retrieval

- Developing a post-hoc nested representation framework for frozen vision–language embeddings, enabling retrieval at multiple dimensionality budgets without re-encoding the gallery.
- Combining modality-specific projection heads, multi-width representation learning, and ranking distillation with adaptive retrieval and calibrated destructive-error risk control.
- Studying retrieval quality, representation memory, inference latency, and efficiency–quality trade-offs across embedding budgets and retrieval settings.
- Manuscript in preparation for *ACM SIGIR 2027*.

Privacy-Preserving Synthetic Data for Agentic Tool-Use Trajectories

- Research project with Prof. Sai Praneeth Karimireddy at FoRT-ML investigating differentially private synthetic data generation for multi-turn agentic tool-use workflows.
- Exploring structural representations of agent trajectories and privacy-preserving generation methods that maintain tool-use coherence and compatibility with executable environments.
- Investigating privacy–utility trade-offs through trajectory validity, replay executability, and downstream agent learning performance.
- Manuscript in preparation, targeting *ICML 2027*.

Technical Skills

Programming: Python, C++, C, Java, R, SQL, MATLAB

Frameworks and Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, Ultralytics

Systems and Tools: Linux, Git, CUDA, LaTeX

Notable Courses

CSCI - 566 : Database Systems (Fall 2025)
CSCI - 585 : Deep Learning and Its Applications (Fall 2025)
CSCI - 677 : Advanced Computer Vision (Fall 2026)
CSCI - 570 : Analysis of Algorithms (Spring 2026)
CSCI - 580 : 3-D Graphics and Rendering (Spring 2026)
CSCI - 618 : Trustworthy Machine Learning from an Optimization Lens (Fall 2026)

Exam Scores

GRE: 325/340 (Quantitative Reasoning: 168/170, Verbal Reasoning: 157/170, Analytical Writing: 4.5)

IELTS: 8.5/9 (Reading: 9, Listening: 9, Speaking: 8, Writing: 7.5)